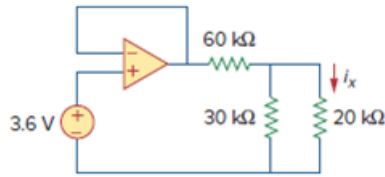


Problem

In the circuit shown in Fig. 5.68, find i_x and the power absorbed by the 20-k Ω resistor.

Figure 5.68:



Step-by-step solution

Step 1 of 4

Consider the following circuit:

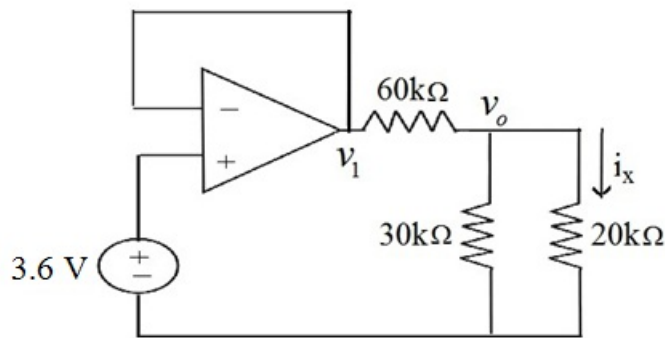


Figure 1

Step 2 of 4

The voltage at the inverting and non-inverting terminals of an ideal op-amp is same. Since, both the terminals are virtually grounded.

Therefore,

$$v_1 = 3.6 \text{ V}$$

Apply KCL at node v_o .

$$\frac{v_o - v_1}{60 \times 10^3} + \frac{v_o}{30 \times 10^3} + \frac{v_o}{20 \times 10^3} = 0$$

$$\frac{v_o - 3.6}{6} + \frac{v_o}{3} + \frac{v_o}{2} = 0$$

$$v_o \left[\frac{1}{6} + \frac{1}{3} + \frac{1}{2} \right] = \frac{3.6}{6}$$

$$v_o = 0.6 \text{ V}$$

Step 3 of 4

Calculate the current i_x .

Apply the Ohm's law.

$$i_x = \frac{v_o}{20 \times 10^3}$$

Substitute 0.6 for v_o in the equation.

$$\begin{aligned} i_x &= \frac{0.6}{20 \times 10^3} \\ &= 30 \mu\text{A} \end{aligned}$$

Therefore, the value of current i_x is $30 \mu\text{A}$.

Step 4 of 4

The power absorbed by the $20 \text{ k}\Omega$ resistor is,

$$P = (i_x)^2 (20 \times 10^3)$$

Substitute 30×10^{-6} for i_x in the equation.

$$\begin{aligned} P &= (30 \times 10^{-6})^2 (20 \times 10^3) \\ &= 18 \mu\text{W} \end{aligned}$$

Therefore, the power absorbed by the $20 \text{ k}\Omega$ resistor is $18 \mu\text{W}$.